General Science

Alveary

About the Course

This is the second of a two-year Physical Science course (i.e., introductory physics and chemistry). The fullness of the course includes topics in earth science, history, politics, and more. The completion of this course provides a gentle yet robust transition into the high school disciplines. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 8

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context, and with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 8
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 8
In level 8, learners study science in its historical, political, and cultural context with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

General Science: Grade 8
For eighth-grade students or possibly hungry seventh graders or ninth graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
8	General Science: Grade 8 3 times/week 30 min	The Story of Science: Einstein Adds a New Dimension The Story of Science: Newton At the Center

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 8				
Natural History: Grade 8	General Science: Grade 8 Nature Walks & Scouting: Grades 1-8	General Science: Grade 8	General Science: Grade 8 Nature Notebook: Grade 8	Labs: Grade 8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 8
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 8
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

General Science: Grade 8
☐ Term 1: A science museum with motion exhibits
☐ Term 2: A science museum with chemistry exhibits
☐ Term 3: A planetarium or local astronomy club event

Term Prep & Teacher Tips

General Science: Grade 8
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
• 5 Starburst candies or similarly-shaped manufactured item
• sturdy wooden board at least 2-3" wide and 3.5' long to construct marble ramp

• anything to create rails for the ramp, more scrap wood, old pool noodles, etc.
• whatever is needed to attach the rails: bar clamps, hammer and nails, glue, etc.
• books or wood blocks to adjust height
• 3 smaller blocks to act as stops
• 1-6 eggs (Term 1)
• water
• large sheet pan or flat location outside
• 2 plastic cups, 1 smaller and 1 larger
• various objects to test for sliding on ramp, such as small objects made of wood, plastic, metal, etc
• various Lego-type blocks that can be used to build a small box, roughly 1-2" dimensions
• a larger bouncy ball, such as a basketball
• a location to bounce balls
• 1/2 c small solid fill material, such as cat food or rice
• 2 locations at 2 different temperatures, such as a sunny window and a chilly basement
• plastic wrap
• wire cutters, such as those on many pliers
• hammer
• at least 2 different light sources, including bright sunlight and at least 1 type of artificial light
• microwave
• microwave-safe casserole dish, at least 9"
• marshmallows to cover the bottom of the dish, Peeps or jumbo work best
• stopwatch, such as found on most electronic devices
• calculator, such as found on most electronic devices
• printables
• computer
• optional: hot glue gun
• optional: long forceps
• optional: matches
• optional: glass jar that is taller than a tea light candle
• optional: hard boiled egg (Term 2)

Reminders

General Science: Grade 8
☐ Note that Lab in week 6 of Term 1 calls for fresh eggs and Lab in week 2 of Term 2 includes an optional activity that uses a hard-boiled egg. Make a note on your calendar if you will need to purchase these closer to the appropriate dates. Refer to the lab book for more details.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

General Science: Grade 8


The Story of Science: Einstein Adds a New Dimension


The Story of Science: Newton At the Center

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ View Basic Supplies
∞ View Supply List Details

General Science: Grade 8

- Household Items - General Science: Grade 8

Quick Links

General Science: Grade 8

- ∞ AmScope key for slide IDs

Click THIS text or scan the QR code for links.

General Science: Grade 8

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

General Science: Grade 8

Click THIS text or scan the QR code for links.

Term 1

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 1**

Setting the Stage

Materials: Newton at the Center

PREP: Read Teacher Tip

→ INTRO
This book tells the story of how physical science developed from the Renaissance into the 1800s. As you move through the story, notice how ideas and knowledge about the world around us changed through a combination of imagination and experimentation. There are many interesting sidebars and tangents in the text, but read the main text first and then explore them, as desired. Watch the course introduction video:
∞ Video Link: Form 3 Science Intro (5:04)

→ READ, NARRATE, & DISCUSS
Ch.1 p.1-6 "For centuries" - "good place to start."

• What do you gather so far about the author's purpose for this chapter?

★ TEACHER TIP
Watch the course introduction video with your student(s).

★ TEACHER TIP
As we will begin some notebooking next week, help students notice the author's structure and purpose this week during discussion.

• IMPORTANT DATES
Beginning of the Renaissance (c.1453)

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 2**

Setting the Stage

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.1 p.6-13 "Imagine yourself" - "familiar landmarks."

• This chapter sets the scene for Copernicus. Describe that scene.
• Consider the relationship between church and state in Copernicus' time. Was a singular church and state good for either the church or the person? If so, in what ways? In what ways was it bad for both the church and the person? Is a separate church and state bad for either the church or the person? If so, in what ways? In what ways is it good for both?

★ TEACHER NOTE
A simplified contrast between medieval philosophical and theological beliefs and modern scientific beliefs is presented on the p.10 sidebar. Teachers might allow it to pass by the way, engage students in discussion, or skip this one (the larger ongoing picture takes shape over the year).

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 3**

Copernicus

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.2 p.14-20 "Copernicus" - "defines the age."

• Tell about Copernicus' life.
• Describe the shift in ideas observed by the author.

• IMPORTANT DATES
Nicolaus Copernicus (1473-1543)

Term 1

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 4**

Copernicus in the Renaissance

Materials: Newton at the Center

PREP: Read Teacher Tip

➔ INTRO
Recall what you read about Copernicus' life and the Renaissance setting for our story. As you continue reading, begin taking notes, jotting down the ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. If you'd like to see a few examples, watch the Notebooking video:
∞ Video Link: Notebooking? (1:39)

➔ READ, NARRATE, & DISCUSS
Ch.2 p.20-31 "Since we have"- "and the earth."

• Describe the contradictions observed in Renaissance society.
• Discuss the scientific and religious ideas on Copernicus' mind.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. Invite them to share their notebook entries to improve the discussion!

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 5**

Leonardo da Vinci

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.2 p.32-35 "What's to be" - "and the Arno."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Leonardo da Vinci.

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 6**

Copernicus Has a Problem

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.3 p.36-43 "Copernicus is" - "move the earth."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What does the author mean by "thought experiment"?
• Discuss some of Copernicus' questions.
• Explain Copernicus's conflict over his ideas.

• IMPORTANT DATES
Copernicus published On the Revolutions of the Heavenly Spheres (1543).

Term 1

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 7**

Tycho Brahe Observes

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.4 p.44-48 "The movement" - "is going on?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Tycho's early life.
• Explain why Tycho's approach was such an important development in thought.
• Why did people think that "perfect" meant "unchanging"? Discuss the implications of this idea.

• IMPORTANT DATES
Tycho Brahe (1546-1601)

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 8**

Tycho Brahe Observes

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.4 p.48-57 "Only once" - "sending up sprouts."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about this episode in Tycho's life.
• Compare the models of Ptolemy, Copernicus, and Tycho using words, drawings, or diagrams.

★ TEACHER NOTE
The timescale of traveling light is mentioned on p.51: "It has taken 20,000 years..." Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

• IMPORTANT DATES
Newton At the Center
Choose events from the timeline on p.56-57.

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 9**

What is Parallax?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.4 p.58-61 "When Tycho" - "a solar system."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain what parallax is and how it is used.

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 10**

Enter Galileo

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

• IMPORTANT DATES
Galileo (1564-1642)

→ READ, NARRATE, & DISCUSS
Ch.5 p.62-66 "Twenty-one years" - "in scientific circles."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Galileo's family and early life.
• Describe the personality traits the author attributes to Galileo. How are these similar or different from the other scientists you've read about?
• Galileo understood the need for "accurate, repeated experiments." Explain how his observation of the swinging chandelier helps him realize this need.

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 11**

Enter Galileo

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.5 p.66-71 "It is in Pisa" - "scientific questioning."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe Galileo's supposed experiment in Pisa.
• How is Galileo's personality getting in his way?
• A recurring theme, the authorities are very concerned about dangerous ideas. What are they afraid of?

★ TEACHER NOTE
The theological beliefs of Bruno are mentioned on p.71 within the context of heresy and politics. Teachers might choose to allow it to pass by the way or to engage students in discussion.

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 12**

Galileo, the Thinker

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.6 p.72-79 "Galileo isn't" - "understand that."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how scientists, like Galileo, use math to understand relationships.
• What are mathematical relationships? (Mathematical ideas, like geometry and patterns, are important!)
• Tell how Galileo discovered his Law of Freefall.

→ SUPPLEMENTAL
∞ Video Link: Hammer Versus Feather on the Moon (1:22)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

General Science: Grade 8

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 5 ☐ **30m General Science: Grade 8 - Lesson 13**

Thinking about Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.7 p.80-83 "Motion. How is" - "because it's important."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain Galileo's thought experiment and his Principle of Relativity.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 5 ☐ **30m General Science: Grade 8 - Lesson 14**

Thinking about Motion

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.7 p.83-89 "What Galileo does" - "at its core."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how Galileo's Relativity applies to our observations, looking at the sky, a moving vehicle, or a thrown ball.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in motion? More interest/ease in discussing the relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a play ground, a science museum with an engineering exhibit, or an art museum with kinetic art. Reach out through Contact Us if you need help.

WEEK 5 ☐ **30m General Science: Grade 8 - Lesson 15**

Thinking about Stars

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.8 p.90-94 "Despite his enemies" - "feel the pull."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Contrast the reaction of most people when they see the supernova or the comet with Galileo's. How is this an echo of Galileo's larger problem with the authorities?
• Describe Galileo's viewpoint on revealed knowledge.

→ SUPPLEMENTAL
∞ Video Link: What is a Comet Made of? (3:31)
∞ Video Link: Night time Lapse of Comet Lovejoy (1:44)

→ NOTE

★ TEACHER NOTE
Galileo distinguishes between answers found through the use of our senses or in the Bible on p.92: "I do not" - "in the Bible." The age of the universe is mentioned throughout p.95-97. Teachers might choose to allow these to pass by the way, to discuss with students, or to consider alternative theories they would like to share. The optional reading could be skipped if preferred.

Read About Suns and Stuff p.95-97 for interest, if desired.
• Describe or diagram the life cycle of a star, including descriptions of each phase.

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 16**

Galileo Challenges the World

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.9 p.98-105 "Galileo is" - "the heavens go."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What happens when Galileo gets a telescope?
• How does Galileo's personality contribute to his conflict with authority?

→ SUPPLEMENTAL
∞ Video Link: Survive a Trip to the Sun (3:20)

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

• IMPORTANT DATES
Invention of the telescope (1609)

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 17**

The Telescope

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.9 p.106-109 "Leonard Digges" - "in the shadows."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Who invented the telescope?

★ TEACHER NOTE
The scale of the universe in millions of light-years is mentioned on p.108. Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 18**

Galileo in Trouble

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.10 p.110-115 "Galileo doesn't" - "of his time."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.10 p.116-117 "Religion is" - "search a bit."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

★ TEACHER TIP
When passages have different main topics, some students will be able to read them through and organize the content in their minds. For other students, it can be helpful to read and narrate the passages separately, even if they are doing the work consecutively in the same lesson period. This helps the mind work through the ideas in smaller chunks.

★ TEACHER NOTE
The balance between

• We see religious leaders both angry with and engaging with Galileo. Why were they ultimately upset with Galileo? How could the situation have been handled differently?
• Explain religion and science using a metaphor of 2 parallel lines of thinking in the mind.
• Is the title of the passage on p.116-117 the best title for the ideas presented? If so, explain. If not, what would you title it?

doctrine and politics is considered on p.113, 116-117. Teachers are encouraged to preread these sidebars for better discussion.

WEEK 7 ⬚ **30m General Science: Grade 8 - Lesson 19**

Kepler's Theories on Planetary Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.11 p.118-124 "Do you think" - "published in 1611."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Kepler.
• Describe Keplar's interest in optics and how this leads to a connection with Galileo.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Johannes Kepler (1571-1630)

WEEK 7 ⬚ **30m General Science: Grade 8 - Lesson 20**

Kepler's Theories on Planetary Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.11 p.124-128 "Meanwhile" - "do just that"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.11 p.129-131 "An ellipse" - "Earth and Sun."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain Kepler's belief that his work as a scientist is God's intent for him, his vocation.
• Contrast Kepler's thoughts on creation with those of previous scientists.
• How does Kepler's work generate new questions?
• What is an ellipse? What is eccentricity?

★ TEACHER NOTE
The timescale of change in Earth's orbital path is mentioned on p.131: "in a 100,000-year cycle." Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

WEEK 7 ⬚ **30m General Science: Grade 8 - Lesson 21**

Descartes and His Coordinates

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

• IMPORTANT DATES
René Descartes (1596-1650)

Ch.12 p.132-136 "Galileo is 32" - "those coordinates."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how the Cartesian coordinate system works.

WEEK 8 ⬚ **30m General Science: Grade 8 - Lesson 22**

Descartes and His Coordinates

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.12 p.136-139 "Like him or not" - "home to rest."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.12 p.141-143 "On June 21" - "hundreds of years."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how even wrong ideas, such as Descartes' vortices, can be helpful in science.
• Tell the story of Fermat's Last Theorem.

WEEK 8 ⬚ **30m General Science: Grade 8 - Lesson 23**

Meet Isaac Newton

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.13 p.144-147 "It turns out" - "ever happening."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Newton's early life.

• IMPORTANT DATES
Isaac Newton (1642-1727)

WEEK 8 ⬚ **30m General Science: Grade 8 - Lesson 24**

Newton Ponders Motion and Math

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.13 p.147-153 "When he is 16" - "he is doing."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe Newton's personality traits. How does he relate to others? What kinds of things does he think about?

Term 1

- Tell the story of how Newton learns math and invents calculus.
- Explain the links in the chain that led to Newton's ideas on universal gravity.

WEEK 9

⬚ 30m General Science: Grade 8 - Lesson 25

Newton Puzzles Over Gravity

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.14 p.154-159 "The celebrated" - "that it doesn't."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Why is this chapter so titled?

WEEK 9

⬚ 30m General Science: Grade 8 - Lesson 26

What is Calculus?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.14 p.160-163 "With arithmetic" - "at any point."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain what calculus is and what it can be used for.

WEEK 9

⬚ 30m General Science: Grade 8 - Lesson 27

Thinking about Light

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.15 p.164-171 "Newton keeps" - "that new world."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Discuss Newton's work related to light.
- Discuss Newton and Hooke's disagreement and their relationship.
- Discuss Newton's concerns about religion.

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 28

Newton's Laws of Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

- **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

Term 1

→ READ, NARRATE, & DISCUSS
Ch.16 p.172-177 "Is motion" - "in his head."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain Newton's Laws of Motion.
• Discuss the implications of these laws in relation to astronomy.

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 29

Fame Finds Newton

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.17 p.178-183 "Newton has" - "himself a star."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss Newton and Halley's relationship. What impact does it have on Newton?
• Why was the Principia important?

• IMPORTANT DATES
Newton publishes Principia (1687)

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 30

Fame Finds Newton

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.17 p.184-187 "The French had" - "last by science."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe the appearance of Halley's Comet from the perspective of someone in 1758. What do you imagine they were thinking?

• IMPORTANT DATES
Halley's Comet (1758)
Appeared last (1986)
Next appearance (2061)

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 31

Measuring Light

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.18 p.188-195 "Can the speed" - "at the sky."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell the story of Roemer and Jupiter's moons.
• Consider the significance of Roemer's realization.

• IMPORTANT DATES
Olaus Christensen Roemer (1644-1710)
Christiaan Huygens (1629-1695)

Term 1

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 32

Measuring Light

Materials: Newton at the Center

PREP: Read Teacher Tip

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.18 p.196-197 "Of the four" - "more impressive."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Study the diagram on p.197. Why would Jupiter's light be brighter sometimes than at other times? If one year equals one trip around the sun, how old would you be if you lived on Jupiter?

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether motion or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help.

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 33

Catch Up

Materials: Newton at the Center

➔ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

WEEK 12

⬚ 30m General Science: Grade 8 - Lesson 34

Exam Week

➔ EXAMS
Answer question(s) related to the course.

Questions will come from:
Newton at the Center

WEEK 12

⬚ 30m General Science: Grade 8 - Lesson 35

Exam Week

➔ EXAMS
Answer question(s) related to the course.

Questions will come from:
Newton at the Center

WEEK 12

⬚ 30m General Science: Grade 8 - Lesson 36

Exam Week

➔ EXAMS
Answer question(s) related to the course.

Questions will come from:
Newton at the Center

Term 2

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 37**

Thinking about Matter

Materials: Newton at the Center

PREP: Read Teacher Tip

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.19 p.198-203 "Aristotle thought" - "close to finding it."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Set the stage for alchemy.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 38**

Thinking about Matter

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ & NARRATE
Ch.19 p.203-207 "Bernard of Treves" - "resounding 'Yes.'"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

➔ READ, NARRATE, & DISCUSS
Ch.19 p.208-209 "Hennig Brandt" - "specific discoverer."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about the demise of alchemy.
• Tell about the discovery of phosphorus.

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 39**

Measuring Matter

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.20 p.210-215 "Irish-born Robert" - "corpuscles seriously."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Robert Boyle.

• IMPORTANT DATES
Robert Boyle (1627-1691)

Term 2

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 40**

Is Air Something - or Nothing?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.20 p.216-219 "Air seems" - "popular entertainment."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Is air something or nothing?

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 41**

Thinking about Energy and Matter

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.21 p.220-225 "Daniel Bernoulli" - "ahead of the times."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about the Bernoulli brothers.

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 42**

Thinking about Energy and Matter

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.21 p.225-229 "Now, let's skip" - "to technology."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain or diagram what you understand about Daniel Bernoulli's principle.

• IMPORTANT DATES
Daniel Bernoulli (1700-1782)

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 43**

"Nature is squaring itself."

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.22 p.230-237 "Think gorgeous" - "built on it."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

★ TEACHER NOTE
The affair between Émilie du Châtelet and Voltaire is introduced on p.232. Teachers might choose to allow it to pass by the way or to discuss with students.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

Term 2

• Tell why Émilie du Châtelet didn't fit in.
• Explain the mistake that Émilie discovered.

• IMPORTANT DATES
Emilie du Châtelet (1706-1749)
Voltaire (1694-1778)

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 44**

Many Gases

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.23 p.238-245 "Just what is" - "have them?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Chemistry is just getting started, but it is very disorganized. Discuss.

★ TEACHER NOTE
The age of some rock is mentioned in a caption on p.239, "roughly 65 million years or so ago." Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

• IMPORTANT DATES
Joseph Priestley (1733-1804)
Benjamin Franklin (1706-1790)

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 45**

The Thermometer

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.23 p.246-249 "Daniel Fahrenheit" - "the temperature."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Daniel Fahrenheit.

• IMPORTANT DATES
Daniel Fahrenheit (1686-1736)

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 46**

Weighing the World

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.24 p.250-259 "Henry Cavendish" - "Earth's density."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how Cavendish measured the density of the world using words, drawings, or diagrams.

★ TEACHER NOTE
The Big Bang is mentioned toward the end of p.258. Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Henry Cavendish (1731-1810)

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 47**

Chemistry is Finally Here!

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.25 p.260-265 "It is the" - "revolution in science."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss what the "big idea" is in this chapter. What is Lavoisier doing?

• IMPORTANT DATES
Antoine-Laurent Lavoisier (1743-1794)

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 48**

Lavoisier Establishes Chemistry

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.26 p.266-274 "His parents" - "ideas survive."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Why is Lavoisier called the "Father of Chemistry"?
• What is an element?
• What do we mean by "conservation of mass"?
• Why is hypothesizing an important part of the scientific method?

→ NOTE
Read France Sings a Metric Tune p.275-277 for interest, if desired.

WEEK 5 ⬚ **30m General Science: Grade 8 - Lesson 49**

Elements and Atoms

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.27 p.278-287 "Way back" - "explain their world."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss how Dalton builds on the ideas of Lavoisier.
• Create a graphic or diagram of Dalton's ideas about elements and atoms.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
John Dalton (1766-1844)

General Science: Grade 8

Term 2

WEEK 5 ⬚ **30m General Science: Grade 8 - Lesson 50**

Atoms and Molecules

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.28 p.288-293 "Amedeo Avogadro" - "complicated things."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.28 p.294-297 "So what makes" - "New Dimension."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain atoms versus molecules.
• Explain Avogadro's Law.
• Discuss how difficult it can be to believe something you cannot see. Consider how people of science and religion both experience this.
• Tell how and what Edward Frankland discovered about chemical bonding.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in chemistry or matter? More interest/ease in discussing the relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a chemistry exhibit. Reach out through Contact Us if you need help.

• IMPORTANT DATES
Amedeo Avogadro (1776-1856)

WEEK 5 ⬚ **30m General Science: Grade 8 - Lesson 51**

Making Sense of Matter

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.29 p.298-308 "Scientists don't" - "that's exciting."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Mendeleyev's life.
• Tell about Mendeleyev's work.

• IMPORTANT DATES
Dmitri Mendeleyev (1834-1907)
Periodic Table of Elements (1869)

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 52**

The Periodic Table

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.29 p.309-311 "So far" - "New Dimension."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell what you know about the Periodic Table.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

• IMPORTANT DATES
The four newest elements added to the Periodic Table (2019)

Click THIS text or scan the QR code for links.

Term 2

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 53**

Thinking about Heat

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.30 p.312-321 "Fill a teacup" - "two philosophers!"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Benjamin Thompson's life.
• Explain Thompson's ideas about heat.

• IMPORTANT DATES
Benjamin Thompson (1753-1814)

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 54**

Studying Electricity

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.31 p.322-327 "You may remember" - "Newton's ideas."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell a story about the Leyden jar.
• How does Benjamin Franklin "link action in the heavens to that on Earth"?

• IMPORTANT DATES
Benjamin Franklin (1706-1790)

WEEK 7 ⬚ **30m General Science: Grade 8 - Lesson 55**

Studying Electricity

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.31 p.327-333 "Franklin isn't" - "path of science."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss how Volta harnesses electricity.
• Explain what is learned about elements and magnetism through this harnessed electricity.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Alessandro Volta (1745-1827)

WEEK 7 ⬚ **30m General Science: Grade 8 - Lesson 56**

A Pendulum's Proof

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

• IMPORTANT DATES
Jean-Bernard-Leon Foucault (1819-1868)
Foucault's Pendulum Experiment (1851)

Term 2

Ch.31 p.334-337 "Way back" - "Amir D. Aczel."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe Foucault's pendulum experiment.

→ SUPPLEMENTAL
∞ Video Link: Foucault's Pendulum Explained (3:20)
∞ Video Link: Foucault Pendulum Time Lapse (watch to 4:45)

WEEK 7 ⬚ **30m General Science: Grade 8 - Lesson 57**

Faraday Experiments

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.32 p.338-345 "It is January" - "he isn't finished."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain Faraday's concept of "fields."
• Tell about Faraday's efforts to improve communication in science. Why is communication so important?
• Discuss Faraday's habit of finding an opportunity to learn in every situation, that is, learning for the sake of learning.
• Explain Faraday's hypothesis and experiment about electricity and magnetism.

• IMPORTANT DATES
Michael Faraday (1791-1867)

WEEK 8 ⬚ **30m General Science: Grade 8 - Lesson 58**

Relating Electricity and Magnetism

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.32 p.345-351 "Faraday is"- "Victoria."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• How was imagination important to Faraday's work?
• Explain how Faraday imagined light in his fields.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 8 ⬚ **30m General Science: Grade 8 - Lesson 59**

Thinking about Light

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.32 p.352-355 "The Bible says"- "an understatement."
Note ideas that stand out to you in words, pictures, or diagrams. Record any

questions or connections you make along the way, too.

• Is light a particle or a wave?

WEEK 8 ⬚ **30m General Science: Grade 8 - Lesson 60**

Maxwell's Equations

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.33 p.356-365 "James Clerk Maxwell"- "of all time."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Maxwell's life.
• Explain what Maxwell does with Faraday's ideas on light. How do the two scientists complement each other?
• "Today, James Clerk Maxwell is ranked with the greatest scientists of all time." Tell why this is, using any combination of pictures and/or words.

• IMPORTANT DATES
James Clerk Maxwell (1831-1879)

WEEK 9 ⬚ **30m General Science: Grade 8 - Lesson 61**

What is Frequency?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.33 p.366-369 "My kids tell"- "than light."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain frequency, using any combination of pictures and/or words.

WEEK 9 ⬚ **30m General Science: Grade 8 - Lesson 62**

What is Heat?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.34 p.370-373, 377-379 "Ludwig Eduard"- "related suicide."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Boltzmann's life.
• Describe how Boltzmann built upon Maxwell's ideas.
• Explain the frustrating debate about atoms.

★ TEACHER NOTE
The suicide of Boltzmann is mentioned in the last sentence on p.379. Teachers might choose to allow it to pass by the way or to discuss with students.

• IMPORTANT DATES
Ludwig Boltzmann (1844-1906)

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 9

⬚ 30m General Science: Grade 8 - Lesson 63

Invisible Ideas

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.34 p.374-376 "Gas molecules" - "and direction."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.34 p.380-383 "In this book" - "managed it."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain the relationship between kinetic energy and temperature.
• Summarize what you know about atoms.

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 64

Putting Energy to Work

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.35 p.384-389 "This is a" - "to human effort."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What is work?
• Explain or draw a series of diagrams that show the potential and kinetic energy of a ball being thrown up into the air.

→ NOTE
Read Information-Age Thinking in the Industrial Era p.390-393 for interest, if desired.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
James Joule (1818-1889)

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 65

First Law of Thermodynamics

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.36 p.394-399 "We have now"- "discoveries of physics."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain "conservation of energy."

• IMPORTANT DATES
Julius von Mayer (1814-1878)
Ferdinand von Helmholtz (1821-1894)

Term 2

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 66

Second Law of Thermodynamics

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.37 p.400-405 "The first law" - "be used again."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Entropy can be tricky to understand. Explain it in terms of "energy spread."

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 67

Second Law of Thermodynamics

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.37 p.405-411 "Now if I've" - "a meaningful answer."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain entropy in terms of probability.
• Is there a difference between understanding entropy in terms of energy spread or probability?

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether chemistry, matter, electricity, magnetism, or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help.

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 68

Things We Don't Understand

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.38 p.412-415 "If you want" - "a Nobel Prize."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What does imagination have to do with science?

General Science: Grade 8

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Term 2

WEEK 11

⬚ **30m General Science: Grade 8 - Lesson 69**

Pursuit of Scientific Truth

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.39-40 p.416-431 "Albert Michelson" - "of all time."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• How would you answer someone today who argues that everything there is to know about a scientific field has already been discovered?
• What is more important in science: theory or experimentation?

• IMPORTANT DATES
Antoine-Henri Becquerel (1852-1908)
First X-Rays (1895)

WEEK 12

⬚ **30m General Science: Grade 8 - Lesson 70**

Exam Week

→ EXAMS
Answer question(s) related to the course.

Questions will come from:
Newton at the Center

WEEK 12

⬚ **30m General Science: Grade 8 - Lesson 71**

Exam Week

→ EXAMS
Answer question(s) related to the course.

Questions will come from:
Newton at the Center

WEEK 12

⬚ **30m General Science: Grade 8 - Lesson 72**

Exam Week

→ EXAMS
Answer question(s) related to the course.

Questions will come from:
Newton at the Center

General Science: Grade 8

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Term 3

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 73**

Introducing Einstein and Recapping

Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

→ INTRO

This book continues the story of science from Newton at the Center. We will learn how the ideas and culture of science developed in surprising directions in the modern era, even as the process of science remained the same. These concepts are challenging to grasp and are meant to stretch the mind and engage the imagination. It is not essential that every technical concept be fully understood. Let the lab book guide you on which concepts to focus on. Break the assigned reading into chunks, as needed. Reading and narrating smaller chunks can be helpful even when reading in one sitting. Allowing the reading to spill over into free reading time is also okay. Learning what works for you is an important skill that will help in high school.

→ READ, NARRATE, & DISCUSS

Ch.1-2 p.1-17 "Fifteen-year-old" - "ever seeing one?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss why Albert Einstein had such a hard time in school.
• Consider what it would be like if you could travel on a beam of light.
• Recall some of the historical events in science that have taken place so far. Recall some of the "big ideas" we've encountered.

★ TEACHER TIP
Passages are not presented in parts this term. Observe how students manage them and encourage them to read and narrate the ideas in chunks, as needed.

★ TEACHER NOTE
A review of religious and political conflict is covered on p.10-12. Teachers might choose to allow it to pass by the way or to discuss with students.

• IMPORTANT DATES
Albert Einstein (1879-1955)
Industrial Revolution (1760-1840)
Euclid (4th B.C.)

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 74**

Electricity, Magnetism, and Light

Materials: Einstein Adds a New Dimension

→ RECAP

Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Ch.3-4 p.18-35 "Science has" - "method employed."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Science is the search for connections and patterns. Discuss this statement.
• Describe Michelson and Morley's experiment.

• IMPORTANT DATES
Samuel Morse (1791-1872) sent an electric current from Washington D.C. to Baltimore (1844).
Thomas Edison (1847-1931)
Nikola Tesla (1856-1943)
Any dates from the timeline on p.20-21

WEEK 1 ⬚ **30m General Science: Grade 8 - Lesson 75**

Thinking about Electrons

Materials: Einstein Adds a New Dimension

→ RECAP

Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Ch.5 p.36-49 "Joseph John Thomson" - "skilled experimenter."
Note ideas that stand out to you in words, pictures, or diagrams. Record any

• IMPORTANT DATES
J.J. Thomson (1856-1940) presents his corpuscles of electricity to the Royal Institution (1897).
Max Planck (1858-1947)
Robert Millikan (1869-1953)

questions or connections you make along the way, too.

• Describe the debate surrounding electricity.

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 76**

What are Atoms Anyway?

▯ Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.6-7 p.50-57 "J.J. has two" - "the uranium."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What do scientists think about atoms at this time?
• Tell what you know about Marie Curie's work.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Marie Curie (1867-1934)
Any dates from the timeline on p.57

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 77**

Investigating Atoms

▯ Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.8 p.58-69 "J.J. Thomson's" - "inside the atom."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how work with radioactivity reveals an understanding of atoms.

• IMPORTANT DATES
Marie and Pierre Curie discovered polonium (1898).
Marie Curie isolated radium (1902).
Chernobyl Nuclear Power Plant explosion (1986)

WEEK 2 ⬚ **30m General Science: Grade 8 - Lesson 78**

Theorizing

▯ Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.9-10 p.70-85 "Max Karl Ernst" - "is turned down."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• If Planck didn't believe his equation, why did he develop it? Why did he even think about it if he wasn't sure it was real?
• Discuss what Einstein did with all of the seemingly contradictory information.

★ TEACHER NOTE
A daughter born to Albert Einstein and Mileva Maric before their marriage is mentioned on p.80. Teachers might choose to allow it to pass by the way or to discuss with students.

• IMPORTANT DATES
Planck's constant quantizes energy (1900)
Einstein published his 5 famous papers (1905).

General Science: Grade 8

[Click THIS text or scan the QR code for links.](#)

Term 3

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 79**

Light as Wave and Particle

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.11 p.86-97 "For 10 years" - "to the explanation."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Light is both a wave and a particle. Discuss.

• IMPORTANT DATES
Lord Rayleigh (1842-1919) explains why the sky is blue (1871).

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 80**

Modeling the Unseen

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.12-13 p.98-113 "It is 1827" - "to find out."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe how the role of mathematics in science is developing.
• Describe Rutherford's gold foil experiment.

• IMPORTANT DATES
Ernest Rutherford (1871-1937) conducted his gold foil experiment, revealing the nucleus of the atom (1909).

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 81**

Getting Atoms

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.14 p.114-127 "Rutherford and" - "astrophysics soared."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how the concept of the atom changed from Thomson to Rutherford to Bohr.
• Explain Bohr and Einstein's difference of opinion. How can they be such good friends when they disagree so fundamentally?

★ TEACHER NOTE
The birth of the universe is mentioned in the sidebar on p.125. Teachers might choose to allow it to pass by the way, to discuss with students, or to consider alternative theories they want to share.

• IMPORTANT DATES
Niels Bohr (1885-1962) introduced his model of the atom (1913).

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 82**

Still Shooting Alpha Particles

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

★ TEACHER NOTE
Carbon-14 dating is introduced in the sidebar on p.135. Teachers might choose to allow it to pass by the way, to discuss with students, or to consider alternative theories they want to

Term 3

Ch.15 p.128-135 "In 1916" - "still ahead."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain what nuclear chemistry continues to reveal about the atom.

share.

• IMPORTANT DATES
Francis Aston (1877-1945) invented a mass spectrograph to identify the isotopes of various elements (1919).
Any dates from the timeline on p.130

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 83**

Matter as Particle and Wave

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.16-17 p.136-149 "Niels Bohr" - "of your view."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What is quantum mechanics?
• Discuss Bohr and Einstein's friendship. Was it so great because of or despite their difference of opinion? Why?
• Discuss the development of the wave-particle idea.

★ TEACHER NOTE
The end of Ch.16 is missing from the Kindle edition. It should include: "United States, as they will soon learn."

• IMPORTANT DATES
Arthur Compton (1892-1962) demonstrated the particle nature of light (1923).
Louis-Victor de Broglie (1892-1987) showed mathematically that matter has wave properties (1924).
G.P. Thomson (1892-1975) demonstrated the wave nature of electrons (1927).

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 84**

Uncertainty Principle

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.18 p.150-157 "The more Niels" - "mentor and poet."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Why is the chapter titled so?
• What statistics are necessary? Why does Einstein dislike this?

★ TEACHER NOTE
The end of Ch.18 is missing from the Kindle edition. It should include: "practical formulas that lead to technological marvels. The founders of quantum mechanics became heroes in an atomic and technological age. Niels Bohr is their mentor and poet."

• IMPORTANT DATES
Werner Heisenberg (1901-1976) introduced the Uncertainty Principle (1925).

WEEK 5 ⬚ **30m General Science: Grade 8 - Lesson 85**

Quantum Mechanics

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

• IMPORTANT DATES
Erwin Schrödinger (1887-1961) formulated his wave equation to describe electrons (1926).
Schrödinger describes duality as his famous cat

Term 3

→ READ, NARRATE, & DISCUSS
Ch.19-20 p.158-172 "Erwin Schrödinger" - "goes to war."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain the idea of Schrödinger's cat and what he meant to explain by it.
• How did Dirac fuse wave mechanics and particle mechanics?
• Tell what you know about atoms so far.

(1935).
Paul Dirac (1902-1984),
John Cockcroft, and Ernest Walton completed their linear particle accelerator (1932),

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WEEK 5 ⬚ **30m General Science: Grade 8 - Lesson 86**

Physics and Chemistry

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.20-21 p.172-189 "After the war" - "one way or another."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain why scientists want to accelerate particles and how Einstein's equation is proven.
• Who was Linus Pauling?

• IMPORTANT DATES
Linus Pauling (1901-1994),
Francis Crick, and James Watson discovered the chemical structure of DNA (1953).
Any dates from the timeline on p.185

WEEK 5 ⬚ **30m General Science: Grade 8 - Lesson 87**

Energy and Mass

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.22-23 p.190-203 "Scientific thoughts" - "World War II begins."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain the implications of Einstein's equation.
• Discuss the collision of science with World War II. This is a discussion that should continue for many lessons!

★ TEACHER NOTE
Legal cases regarding evolution in the classroom and victimized/marginalized groups are mentioned on p.194, 198, respectively. Teachers might choose to allow these to pass by the way or to discuss with students.

• IMPORTANT DATES
Any events noted in Ch.23 (the chapter serves as an expanded timeline)

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 88**

The Race to Fission

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.24 p.204-219 "It is September" - "nuclear energy."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain what you know about how nuclear fission of the uranium nucleus occurs.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

• IMPORTANT DATES
Irène and Frédéric Joliot-Curie created the first radioactive isotope artificially (1933).

Term 3

Enrico Fermi split the uranium nucleus (1934).
Lise Meitner and Otto Frisch discovered that nuclear fission had occurred in the Hahn lab (1938).

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 89**

The Bomb

▯ Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.25 p.220-235 "In 1938" - "else possible."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe the scientific race that is beginning.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in energy or light? More interest/ease in discussing the process of science or the role that curiosity and uncertainty play? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a planetarium. Reach out through Contact Us if you need help.

• IMPORTANT DATES
Einstein wrote his letter to FDR (1939).
Glenn Seaborg discovered plutonium (1940).
Self-sustaining nuclear chain reaction (1942)

WEEK 6 ⬚ **30m General Science: Grade 8 - Lesson 90**

The Manhattan Project

▯ Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.26 p.236-253 "In 1943" - "lost its innocence."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell the story of the Manhattan Project from the perspective of the scientists.

★ TEACHER NOTE
The end of Ch.26 is missing from the Kindle edition. It should include: "on making the world better, well, science had lost its innocence."

• IMPORTANT DATES
Hiroshima and Nagasaki were bombed (1945).

WEEK 7 ⬚ **30m General Science: Grade 8 - Lesson 91**

Quantum Electrodynamics

▯ Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Term 3

Ch.27-28 p.254-165 "The theoretical" - "they never have."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• How does science proceed after the war?
• Recall Galileo's relativity and explain Einstein's ideas.
• What is the big idea about QED? How does it relate to previous ideas? How does it improve our model of the universe?

WEEK 7 **30m General Science: Grade 8 - Lesson 92**

Relativity

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.29 p.266-275 "When Isaac" - "you inhabit."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain spacetime.
• Why were previous models of the universe wrong? What key idea was necessary?
• How can stretching the mind and engaging the imagination be helpful in our pursuit of "I AM"?

WEEK 7 **30m General Science: Grade 8 - Lesson 93**

Spacetime

Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.30-31 p.276-289 "Einstein asks" - "he be thinking?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss how different realities can exist for different observers, according to relativity.
• Discuss time from our perspective versus time from the perspective of a beam of light.
• Consider that Einstein's thought was steeped in Jewish theology. Discuss any spiritual implications that come to mind.

★ TEACHER TIP
Don't worry if the ideas in the text don't quite make sense - they are challenging! These are lifetime ideas for most of us. Engaging with them to see the big picture is what's important.

WEEK 8 **30m General Science: Grade 8 - Lesson 94**

Speed of Light

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

Ch.32-33 p.290-301 "Albert Einstein" - "under your hat."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Continue discussing Einstein's challenging idea!

WEEK 8

30m General Science: Grade 8 - Lesson 95

Relative Gravity

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.34 p.302-311 "Gravity. The very" - "with a red flower."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain Einstein's thought experiment regarding gravity.
• What does gravity have to do with relativity?

WEEK 8

30m General Science: Grade 8 - Lesson 96

Spacetime and the Universe

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.35-36 p.312-325 "By 1915" - "surprises are coming."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain gravity in terms of spacetime.
• Explain how a solar eclipse proved Einstein's idea.
• Einstein was not perfect. Discuss his acceptance of this fact based on what you have read.

→ SUPPLEMENTAL
∞ Video Link: General Relativity Demo (3:20)

★ TEACHER NOTE
The end of Einstein's first marriage and his relationship with his second wife are introduced on p.317. Ideas concerning the temporal and spatial finiteness of the universe are discussed on p.322-325. Teachers might choose to allow these to pass by the way or to discuss with students.

• IMPORTANT DATES
General relativity was confirmed during the solar eclipse (1919).

WEEK 9

30m General Science: Grade 8 - Lesson 97

An Expanding Universe

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.37-38 p.326-339 "In 1917" - "thinks it is."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain what is meant by red-shifted light.
• Explain why scientists hypothesized that the universe is expanding.
• Discuss Lemaître's ideas.

★ TEACHER NOTE
Ideas concerning the expansion of the universe and the introduction of the Big Bang theory are discussed on p.326-339. Teachers might choose to allow this to pass by the way, to discuss with students, or to consider alternative theories they would like to share.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work

Term 3

Quick Link.

• **IMPORTANT DATES**
Edwin Hubble (1889-1953) discovered other galaxies (1925) and observed the expansion of the universe (1929).

WEEK 9 ⬚ **30m General Science: Grade 8 - Lesson 98**

The Fates of Stars

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.39-40 p.340-355 "His real name" - "is at war."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Chandra.
- Tell what you know about stars.

• **IMPORTANT DATES**
Subrahmanyan Chandrasekhar (1910-1995) discovered that certain stars do not become white dwarfs.
Any dates from the timeline on p.352

WEEK 9 ⬚ **30m General Science: Grade 8 - Lesson 99**

Black Holes

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.41 p.356-369 "When supermassive" - "gets his magazine."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell what you know about black holes.

★ TEACHER NOTE
The Big Bang is considered in the quote on p.359; a bet with a payoff of a "subscription to a racy magazine" is mentioned on p.365; the age of the Earth and the Big Bang are mentioned in the sidebar on p.368. Teachers might choose to allow these to pass by the way or to discuss with students.

• **IMPORTANT DATES**
Astronauts Armstrong, Aldrin, and Collins landed on the moon (1969).

WEEK 10 ⬚ **30m General Science: Grade 8 - Lesson 100**

Gravity Waves?

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.42 p.370-381 "Deep in a pine" - "Stay tuned."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

★ TEACHER NOTE
The Big Bang and the age of the universe are further considered on p.372-373, 377. Teachers might choose to allow this to pass by the way or to discuss with students.

• Describe the 4 interactions between matter: strong, electromagnetism, weak, and gravity.

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 101

More Unseen in Space and Time

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.43 p.382-391 "It is 1948" - "with such precision."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What do particles and waves have to do with the "big bang"?

★ TEACHER NOTE
The Big Bang and the age of the universe are further considered on p.382-391. Teachers might choose to allow this to pass by the way or to discuss with students.

WEEK 10

⬚ 30m General Science: Grade 8 - Lesson 102

Inflation and the Theory of Everything

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.44 p.392-405 "A tiny acorn" - "cosmological system."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What is "inflation" to an astrophysicist?

→ NOTE
Read Ch.45 p.406-413 for interest, if desired.

★ TEACHER NOTE
The Big Bang and the age of the universe are further considered on p.392-405 and in the optional reading on p.408. Teachers might choose to allow this to pass by the way or to discuss with students.

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 103

Difficult Predictions

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.46 p.414-427 "Brian Schmidt" - "be the limit."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss the problem with the theory of expansion.

★ TEACHER NOTE
The Big Bang and the age of the universe are further considered on p.415-425. Teachers might choose to allow this to pass by the way or to discuss with students.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 104

Difficult Predictions

Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether energy, light, the process of science, or the

Term 3

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.47 p.428-437 "Clocks were" - "it's so long!"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss how we use metaphors to help us think about ideas.

habits we need to practice as scientists are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help or have any concerns about readiness for the next stage in relationship building.

WEEK 11

⬚ 30m General Science: Grade 8 - Lesson 105

Difficult Predictions

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.48-49 p.438-454 "Do you believe" - "yet to discover."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• "The classical scientists believed that understanding nature would bring the power of exact prediction." Discuss.
• Discuss why science is about what we don't know rather than what we do.
• Why is the everyday still complex and difficult to predict, despite all that science has learned?
• Consider Einstein's question printed on p.449: "Did God have any choice in the creation of the world?" If the universe has to be the way that it is for "deep mathematical reasons," does that prohibit God's choice and creativity? Or does that rather tell us something about God?
• Do some research to find out how ideas have changed or developed around this question of life on other planets.

★ TEACHER NOTE
The age of the universe is mentioned on p.448. Teachers might choose to allow this to pass by the way or to discuss with students.

WEEK 12

⬚ 30m General Science: Grade 8 - Lesson 106

Exam Week

➔ EXAMS
Answer question(s) related to the course.

Questions will come from:
Einstein Adds a New Dimension

WEEK 12

⬚ 30m General Science: Grade 8 - Lesson 107

Exam Week

➔ EXAMS
Answer question(s) related to the course.

Term 3

Questions will come from:
Einstein Adds a New Dimension

WEEK 12

⬚ **30m General Science: Grade 8 - Lesson 108**

Exam Week

➔ EXAMS
Answer question(s) related to the course.

Questions will come from:
Einstein Adds a New Dimension

General Science: Grade 8

Examination

Term 1

· Newton at the Center: Explain how the study of motion helped scientists, like Brahe, Galileo, and Kepler, learn new things about planetary movement.

· Newton at the Center: Explain both perspectives of the conflict between Galileo and the Church.

· Newton at the Center: Explain Newton's 3 Laws of motion using words, drawings, and/or diagrams.

· Explain how you did one laboratory investigation this term and what you learned from it.

Term 2

· Newton at the Center: Newton, Boyle, Cavendish, Lavoisier, and many others made important discoveries about matter. Explain how their work helped transition alchemy into a true science, called chemistry.

· Newton at the Center: Explain 3 ways that moving particles can affect other matter.

· Newton at the Center: Discuss the relationship between electricity and magnetism.

· Explain how you did one laboratory investigation this term and what you learned from it.

Term 3

· Einstein Adds a New Dimension: Tell what you know about the nature of light.

· Einstein Adds a New Dimension: Explain what is meant by E = mc^2 and give examples as to how this is relevant in the world.

· Einstein Adds a New Dimension: Describe the process of science, using examples from the text, as needed.

· Explain how you did one laboratory investigation this term and what you learned from it.